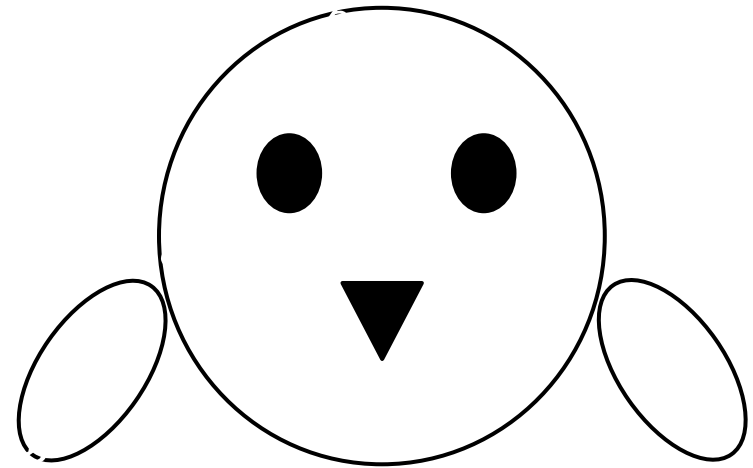


Ghost

Narrative Puzzle Game for Chatbot/NLP Learning

4th Group Meeting · MSc Advanced Computing · King's College London

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Project Goal and Flow

Main question

How can selected IBM SkillsBuild chatbot and NLP concepts be turned into playable puzzle levels in Ghost?

Game-based learning

Actions should directly teach content: the player learns the concept by doing the concept.

Narrative & agents

Ghost and Lily turn abstract NLP topics into a motivated repair task with social scaffolding.

LLM support boundary

Future LLM use is for hints, explanations, and response variation
scoring remains deterministic.

1 Communication problem

Ghost misunderstands or replies out of order.



2 Puzzle action

Player classifies, tags, or connects.



3 Deterministic check

Validators decide correctness, not an LLM.



4 Visible response

Ghost's reaction changes based on the result.

DEMO — Game Shell + Narrative

Ghost

A placeholder shell for Lily's guidance, Ghost's presence, and act selection.

Welcome to Ghost

Help a cute ghost understand messages by repairing little chatbot and NLP puzzles.

Start / Continue

Companions

Ghost

Cute ghost placeholder
Waiting for clearer messages.

Lily

Lab senior placeholder
Nervous, kind, and very prepared.

Lily

Lily

Um... hi. I'm Lily from the lab. Ghost's messages are getting tangled, so maybe we can help it understand what people mean.

What this demonstrates

- Player name entry feeds narrative state.
- Lily dialogue frame introduces each Act.
- Return-to-hub overlay keeps acts connected.

DEMO — Acts 1

Act 1: Intent classification

Players group messages by purpose, then receive deterministic feedback from the validator.

Act 1: Intent Sorting

[Return to Hub](#)

Group messages by speaker intent, not exact wording.
Click or drag a card into an intent group. Drag assigned cards back or between groups to fix mistakes. Click Validate when ready.

Unassigned Messages

I lost my little brass key.

Has anyone seen my notebook?

Can Ghost help me look for the missing lantern?

Where are you floating right now?

Which room are you in?

Are you near the old door?

Who are you?

What should I call this little ghost?

Can you tell me your name?

Intent Groups

find_item

Messages asking Ghost to help find something.

Back: I lost my little brass key.

Back: Has anyone seen my notebook?

Back: Can Ghost help me look for the missing lantern?

ask_location

Messages asking where Ghost is.

Back: Where are you floating right now?

Back: Which room are you in?

Back: Are you near the old door?

ask_identity

Messages asking who Ghost is.

Back: Who are you?

Back: What should I call this little ghost?

Back: Can you tell me your name?

[Validate grouping](#)

All messages are grouped by intent.

Lily

Lily
Sorry, I think out loud when I'm nervous. Ignore me.

[Next](#)

DEMO — Acts 2

Act 2: Entity extraction

Players tag spans and entity types, including system/custom distinction and synonym cases.

Act 2: Entity Extraction

[Return to Hub](#)

Select a word chip, choose the entity type Ghost should notice, then Validate.
Click a tagged chip to untag it and re-check the details.

Message Word Chips

Ghost

heard

humming

in

the

lab

room

at

9pm

time

Validate spans

Tag the important details, then press Validate spans.

Lily

Lily
Look, Ghost's noticing nouns now. They grow up so

Next

Entity Types

time
System

room
Custom

object
Custom

DEMO — Act 3 Node Graph

Act 3: Ghost's Reply Map

Return to Hub

Help Ghost reply in the right order. Add simple cards, move them around, then drag wires between their ports.

Palette

Flow

Start here

Where Ghost

Recogniz

Visitor wants

Check

Check

Does Ghost

Reply

Answer

Use this when

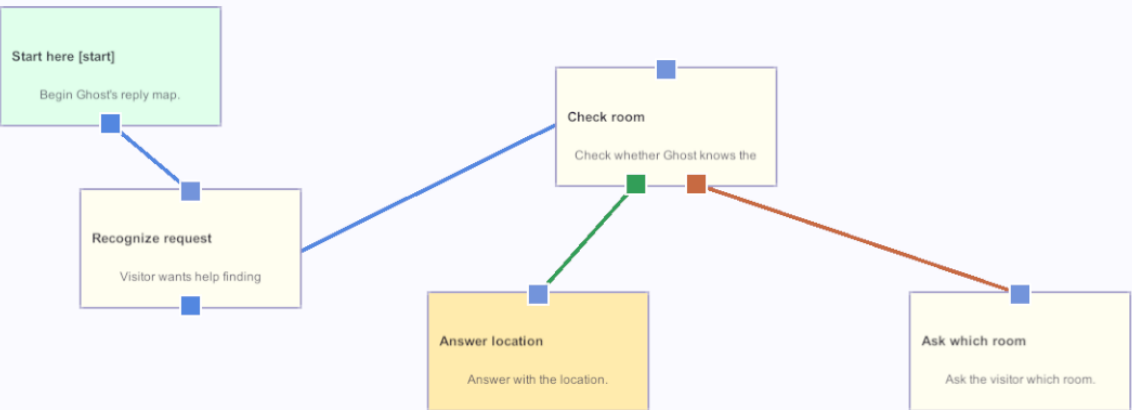
Ask which

Use this when

Reply Map

Goal: If the visitor names a room, Ghost answers. If the room is missing, Ghost asks which room.

Move cards freely. Drag from a bottom port to the next card's top port.



Test Ghost's map Nice. Ghost answers when the room is known and asks when it is missing.

X drop card

Guide

How to play

Drag cards into the map. Connect dots so Ghost checks the room before replying. Drag cards to X to delete. Click a wire or card and press Del to remove it.

Legend

- Blue dot: next step
- Green dot: room is known
- Orange dot: room is missing
- Top dot: drop a wire here

Ghost should handle

Room is known
Visitor asks for the lantern in lab. Ghost should

Room is missing
Visitor asks for the lantern. Ghost should ask which

Ghost reaction

Ghost understands the route: with a room, it answers; without a room, it politely asks which room.

Lily

Lily

Start with where Ghost begins, then give it a path.

Next

Reply-map gameplay

- Drag cards from palette to canvas.
- Connect ports with straight wires.
- Test Ghost's map with target cases.
- Success/failure changes Ghost's reaction.
- Trash / Delete removes nodes or wires.

Current Prototype Status

Backend log proof: attempts and progress saved in SQLite

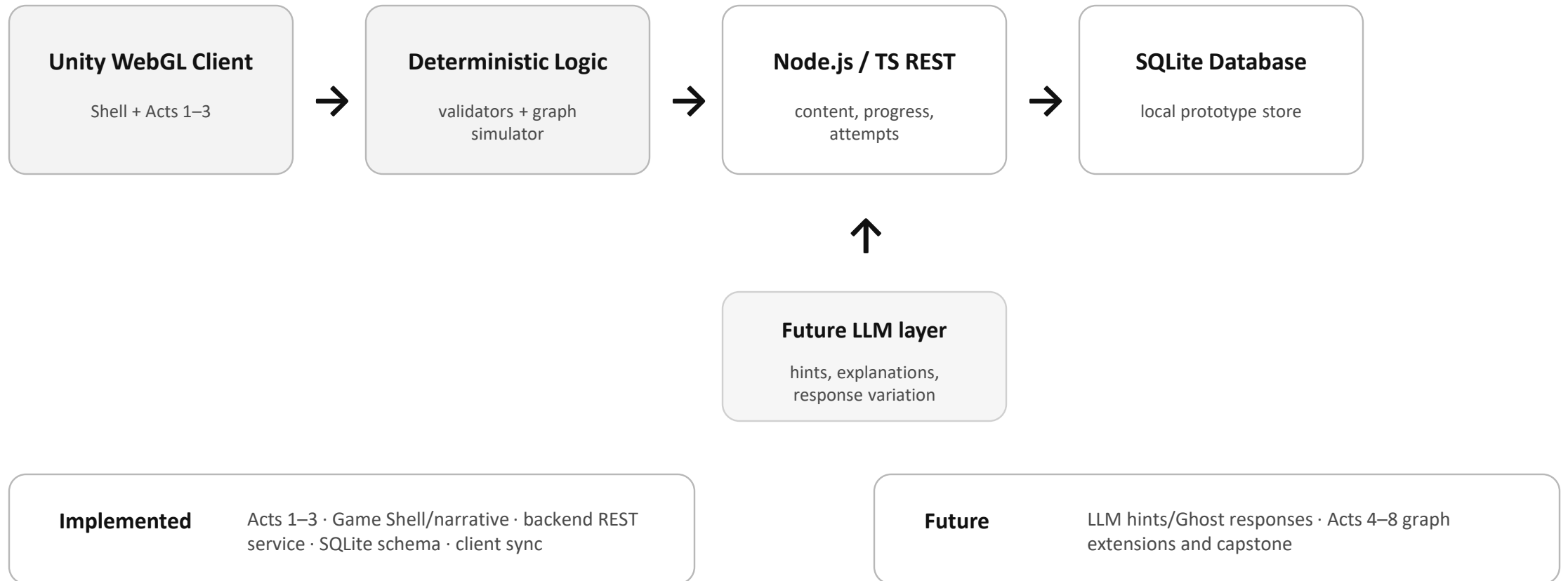
```
PS D:\code\Ghost\Backend> node -e "const Database=require('better-sqlite3'); const db=new Database('data/ghost.sqlite'); console.table(db.prepare('select id, profile_id, act_id, result, created_at from attempts order by id desc limit 20').all()); console.table(db.prepare('select profile_id, acts_completed_json, levels_completed_json, narrative_state_json, up dated_at from progress').all());"
```

(index)	id	profile_id	act_id	result	created_at
0	4	'profile_bc24b0c7-775e-48af-9f40-322fabbbfaaa'	'act1'	'correct'	'2026-06-24T14:25:52.101Z'
1	3	'profile_bc24b0c7-775e-48af-9f40-322fabbbfaaa'	'act3'	'incorrect'	'2026-06-24T14:23:16.013Z'
2	2	'profile_bc24b0c7-775e-48af-9f40-322fabbbfaaa'	'act2'	'correct'	'2026-06-24T14:23:04.294Z'
3	1	'profile_bc24b0c7-775e-48af-9f40-322fabbbfaaa'	'act1'	'incorrect'	'2026-06-24T14:22:30.656Z'

(index)	profile_id	acts_completed_json	levels_completed_json
narrative_state_json	updated_at		
0	'profile_bc24b0c7-775e-48af-9f40-322fabbbfaaa'	'["act1","act2","act3"]'	'["act1:1","act2:1","act3:1"]'
'{"playerName":"Alibaba"}'	'2026-06-24T14:23:24.547Z'		

Area	Results
Playable content	Game shell and Acts 1–3 screens are implemented
Puzzle checking	Rule-based validators are implemented
Story data	Act intro and in-act dialogue data are connected
Backend	Content, profile, progress, and attempt logging APIs are working

System Architecture: Planned vs Implemented



How much LLM integration is necessary for final assessment?